



KUALA LUMPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

(formerly known as Infrastructure University Kuala Lumpur)

ASSIGNMENT 2: 3D GRAPHICS DESIGN PROJECT

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1. Introduction

1.1 Project Overview

For my Computer Graphics (CSC3031) assignment, I was tasked with designing a 3D scene using OpenGL. I chose to design a "**School Classroom**". I selected this theme because it is a familiar environment that requires careful detail to look realistic. A classroom is not just an empty room; it needs organized furniture, specific lighting, and functional elements like doors and fans to feel correct.

1.2 Objectives

My main goal was to move beyond simple shapes. I wanted to build complex furniture by combining basic cubes. I also aimed to master **Hierarchical Modeling**, ensuring that objects like chairs stay attached to their desks when moved. Additionally, I wanted to implement a realistic "Day and Night" cycle to change the atmosphere of the room. Finally, I faced the structural challenge of creating a working door that reveals an actual opening in the wall.

2. Resource Gathering

To complete this assignment, I utilized the following tools:

- **Code::Blocks IDE:** My primary environment for writing and compiling C++.
- **GLUT Library:** I used this for handling window creation and rendering 3D shapes.
- **Mathematics:** I relied on basic geometry (X, Y, Z coordinates) to align the rows of desks perfectly.

3. Design and Implementation

This section explains the code and logic behind the main features of the classroom.

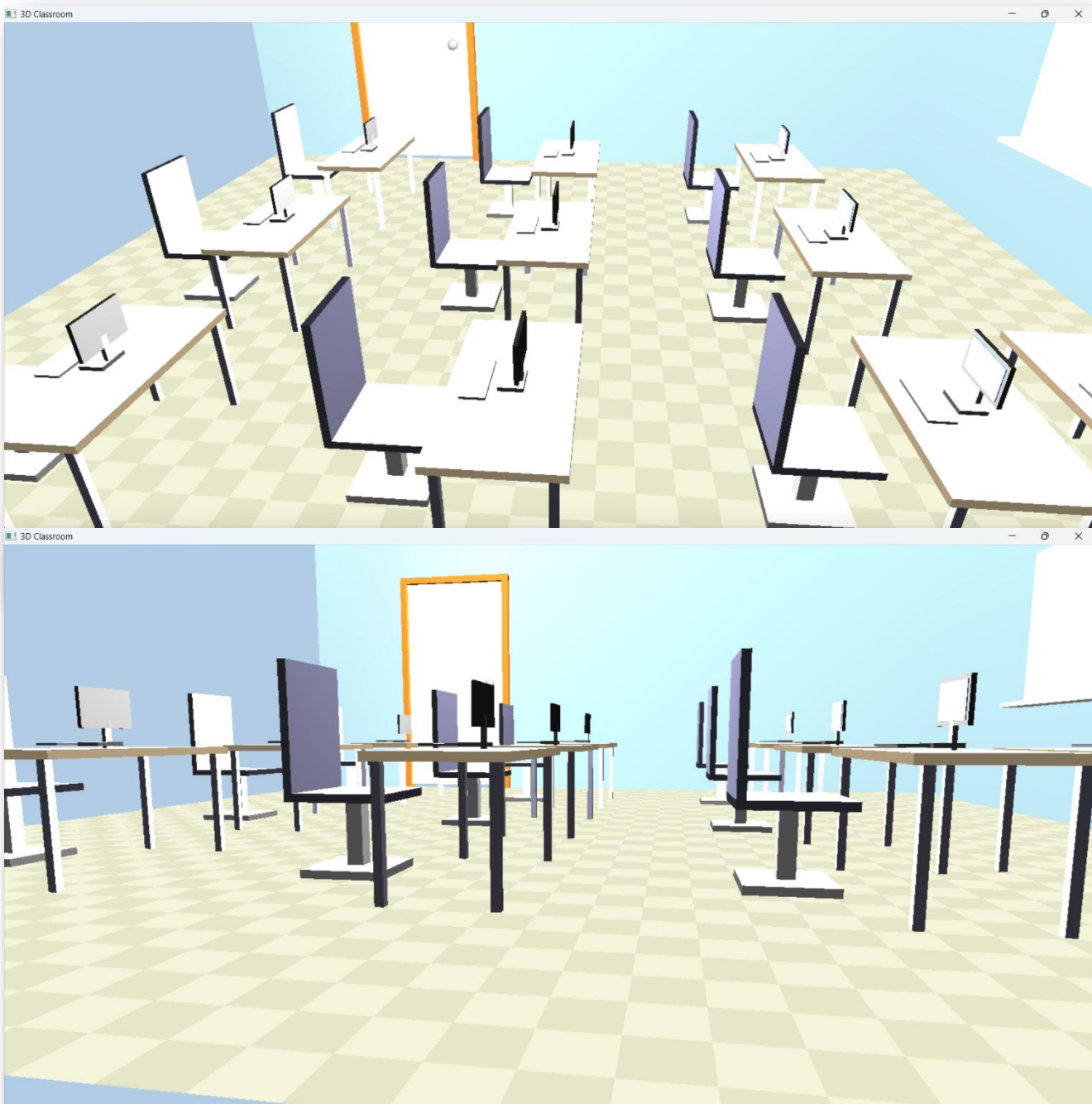
3.1 Hierarchical Modeling (The Student Chair)

In OpenGL, shapes do not naturally stick together. If you draw a seat and legs separately, moving one leaves the others behind. To fix this, I used "Hierarchical Modeling."

I grouped the parts of the chair using `glPushMatrix()`. This command saves the current position. Then, I drew the seat and the legs relative to that center point. This ensures that the entire bench acts as one solid object.

The Code:

```
146 void drawStudentChair() {  
147     glPushMatrix();  
148     glColor3f(0.2f, 0.2f, 0.25f);  
149     glPushMatrix(); glTranslatef(0.0f, 1.0f, 0.0f); glScalef(1.2f, 0.1f, 1.2f); glutSolidCube(1.0f); glPopMatrix();  
150     glPushMatrix(); glTranslatef(0.0f, 1.8f, 0.55f); glScalef(1.2f, 1.5f, 0.1f); glutSolidCube(1.0f); glPopMatrix();  
151     glColor3f(0.5f, 0.5f, 0.5f);  
152     glPushMatrix(); glTranslatef(0.0f, 0.5f, 0.0f); glScalef(0.2f, 1.0f, 0.2f); glutSolidCube(1.0f); glPopMatrix();  
153     glPushMatrix(); glTranslatef(0.0f, 0.1f, 0.0f); glScalef(1.0f, 0.1f, 1.0f); glutSolidCube(1.0f); glPopMatrix();  
154     glPopMatrix();  
155 }  
156
```

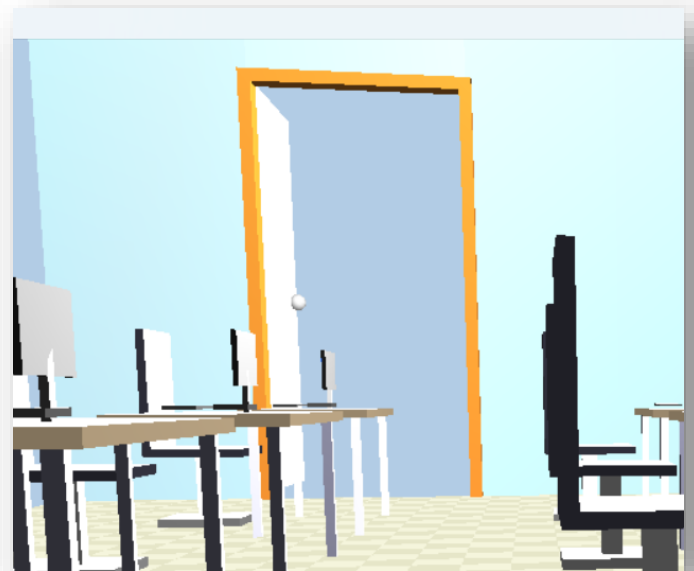
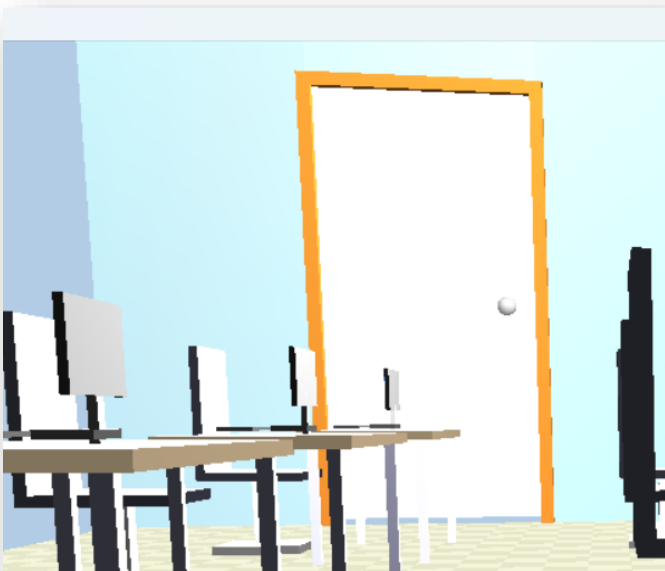


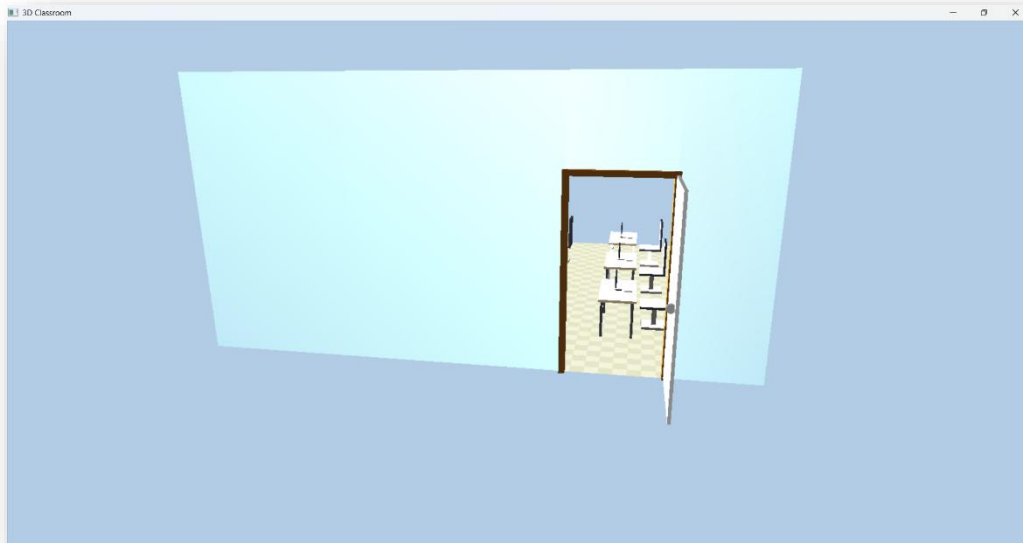
3.2 The Hollow Door Frame

Designing the door was challenging. Initially, drawing a door frame as a solid block blocked the view even when the door was open. To solve this, I modeled the frame as three separate thin pieces: a Left Post, a Right Post, and a Top Header. This creates a physical hole in the center that the user can see through.

The Code:

```
182 // --- HOLLOW DOOR ---
183
184 void drawDoor() {
185     glPushMatrix();
186     glTranslatef(-10.0f, 0.0f, 5.0f);
187
188     // Door Frame
189     glColor3f(0.5f, 0.3f, 0.1f);
190     glPushMatrix(); glTranslatef(0.0f, 3.5f, -1.75f); glScalef(0.4f, 7.0f, 0.2f); glutSolidCube(1.0f); glPopMatrix();
191     glPushMatrix(); glTranslatef(0.0f, 3.5f, 1.75f); glScalef(0.4f, 7.0f, 0.2f); glutSolidCube(1.0f); glPopMatrix();
192     glPushMatrix(); glTranslatef(0.0f, 6.9f, 0.0f); glScalef(0.4f, 0.2f, 3.7f); glutSolidCube(1.0f); glPopMatrix();
193
194     // Animated Panel
195     glPushMatrix();
196     glTranslatef(0.0f, 0.0f, 1.75f);
197     glRotatef(doorAngle, 0.0f, 1.0f, 0.0f);
198     glTranslatef(0.0f, 0.0f, -1.75f);
199
200     glColor3f(0.9f, 0.9f, 0.9f);
201     glPushMatrix(); glTranslatef(0.0f, 3.5f, 0.0f); glScalef(0.1f, 6.8f, 3.3f); glutSolidCube(1.0f); glPopMatrix();
202
203     glColor3f(0.8f, 0.8f, 0.8f);
204     glPushMatrix(); glTranslatef(0.15f, 3.5f, -1.2f); glutSolidSphere(0.15f, 10, 10); glPopMatrix();
205     glPopMatrix();
206     glPopMatrix();
207 }
```





3.3 Adding Realism (Computers on Desks)

A modern classroom looks empty without technology. To add realism, I created a `drawComputer()` function. This function constructs a monitor using a base, a stand, and a screen. I added a small blue cube for the screen to make it look like it is powered on. I placed this object on top of every student desk.

The Code:



```

73 // --- OBJECT: COMPUTER (MONITOR & KEYBOARD) ---
74
75 void drawComputer() {
76     glPushMatrix();
77     // 1. Monitor Base
78     glColor3f(0.2f, 0.2f, 0.2f); // Dark Grey
79     glPushMatrix();
80     glTranslatef(0.0f, 0.05f, 0.0f);
81     glScalef(0.4f, 0.05f, 0.3f);
82     glutSolidCube(1.0f);
83     glPopMatrix();
84
85     // 2. Monitor Stand
86     glPushMatrix();
87     glTranslatef(0.0f, 0.2f, -0.1f);
88     glScalef(0.1f, 0.4f, 0.05f);
89     glutSolidCube(1.0f);
90     glPopMatrix();
91
92     // 3. Monitor Screen Frame
93     glColor3f(0.1f, 0.1f, 0.1f); // Black
94     glPushMatrix();
95     glTranslatef(0.0f, 0.5f, -0.1f);
96     glScalef(0.8f, 0.5f, 0.05f);
97     glutSolidCube(1.0f);
98     glPopMatrix();
99
100    // 4. The Screen (Glowing Blue)
101    if (isLightOn) glColor3f(0.2f, 0.4f, 0.8f); // Windows Blue
102    else glColor3f(0.1f, 0.2f, 0.4f); // Dimmer at night
103
104    glPushMatrix();
105    glTranslatef(0.0f, 0.5f, -0.07f); // Slightly in front of frame
106    glScalef(0.7f, 0.4f, 0.01f);
107    glutSolidCube(1.0f);
108    glPopMatrix();
109
110    // 5. Keyboard
111    glColor3f(0.25f, 0.25f, 0.25f); // Grey
112    glPushMatrix();
113    glTranslatef(0.0f, 0.05f, 0.4f);
114    glScalef(0.8f, 0.03f, 0.3f);
115    glutSolidCube(1.0f);
116    glPopMatrix();
117
118    glPopMatrix();
119 }

```

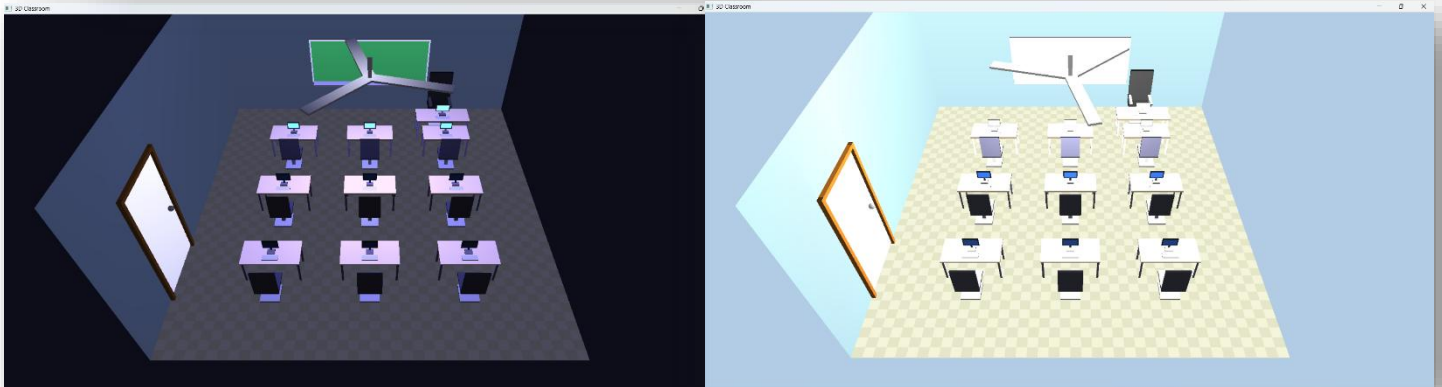
3.4 Interactive Lighting (Day and Night)

Lighting changes the mood of the scene. I implemented a logic system using a boolean variable isLightOn.

- **Day Mode:** The background is sky blue, and the light is bright white.
- **Night Mode:** The background turns black, and the light dims to a dark blue to simulate moonlight.

The Code:

```
221
222 // --- DISPLAY ---
223 void display(void) {
224     if (isLightOn) {
225         glClearColor(0.7f, 0.8f, 0.9f, 1.0f);
226         glEnable(GL_LIGHT0);
227         GLfloat lightPos[] = { 0.0f, 9.0f, 0.0f, 1.0f };
228         GLfloat lightCol[] = { 0.8f, 0.8f, 0.8f, 1.0f };
229         glLightfv(GL_LIGHT0, GL_POSITION, lightPos); glLightfv(GL_LIGHT0, GL_DIFFUSE, lightCol);
230         GLfloat ambient[] = { 0.4f, 0.4f, 0.4f, 1.0f };
231         glLightfv(GL_LIGHT0, GL_AMBIENT, ambient);
232     } else {
233         glClearColor(0.05f, 0.05f, 0.1f, 1.0f);
234         GLfloat lightPos[] = { 0.0f, 9.0f, 0.0f, 1.0f };
235         GLfloat lightCol[] = { 0.1f, 0.1f, 0.2f, 1.0f };
236         glLightfv(GL_LIGHT0, GL_POSITION, lightPos); glLightfv(GL_LIGHT0, GL_DIFFUSE, lightCol);
237         GLfloat ambient[] = { 0.05f, 0.05f, 0.1f, 1.0f };
238         glLightfv(GL_LIGHT0, GL_AMBIENT, ambient);
239     }
240
241     glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);
242     glLoadIdentity();
243
244     gluLookAt(0.0f, 12.0f, cameraZoom, 0.0f, 0.0f, 0.0f, 0.0f, 1.0f, 0.0f);
245     glRotatef(cameraAngleY, 1, 0, 0);
246     glRotatef(cameraAngleX, 0, 1, 0);
247 }
```

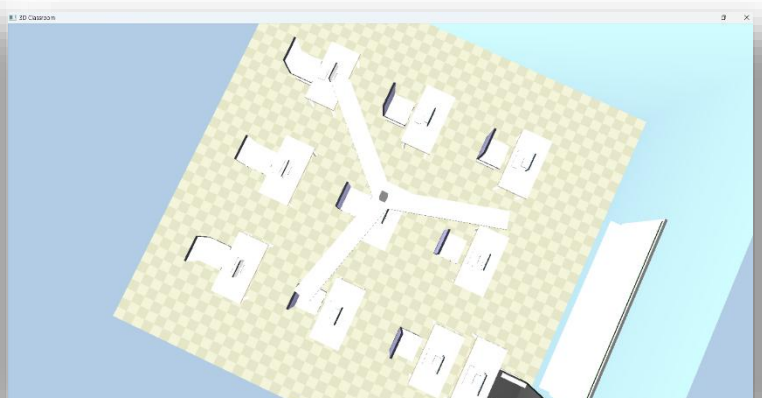
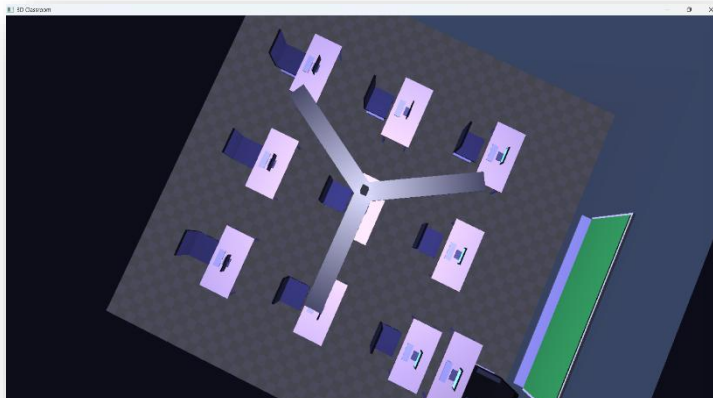
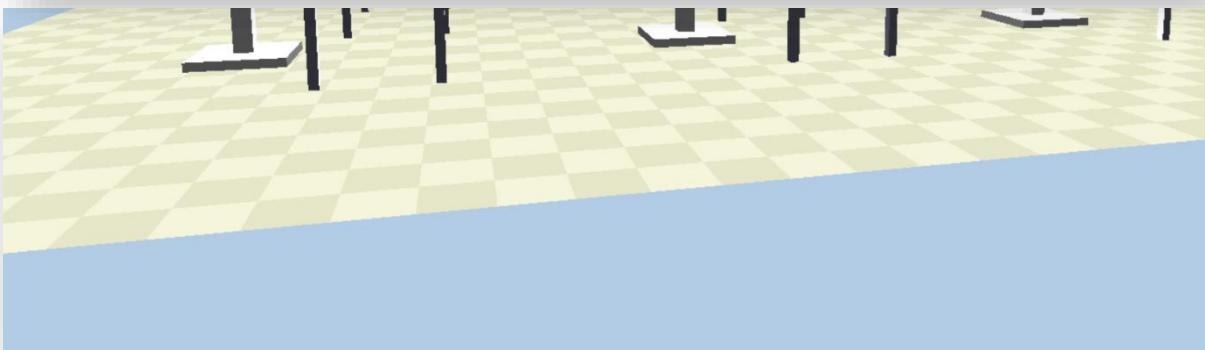


3.5 Procedural Floor Texture

Instead of using an image file, I generated the floor texture using math. I wrote a function that loops through the texture pixels. It uses a bitwise operator (XOR) to decide if a pixel should be cream or beige. This creates a perfect checkerboard tile pattern without needing external files.

The Code:

```
248 // FLOOR
249 glEnable(GL_TEXTURE_2D);
250 glBindTexture(GL_TEXTURE_2D, texName);
251 glColor3f(1.0f, 1.0f, 1.0f);
252 glBegin(GL_QUADS);
253     glNormal3f(0.0f, 1.0f, 0.0f);
254     glTexCoord2f(0.0f, 0.0f); glVertex3f(-10.0f, 0.0f, 10.0f);
255     glTexCoord2f(5.0f, 0.0f); glVertex3f( 10.0f, 0.0f, 10.0f);
256     glTexCoord2f(5.0f, 5.0f); glVertex3f( 10.0f, 10.0f, 10.0f);
257     glTexCoord2f(0.0f, 5.0f); glVertex3f(-10.0f, 10.0f, 10.0f);
258 glEnd();
259 glDisable(GL_TEXTURE_2D);
260
```

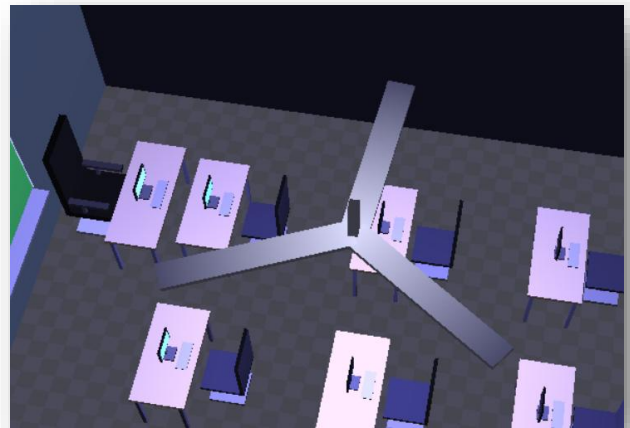
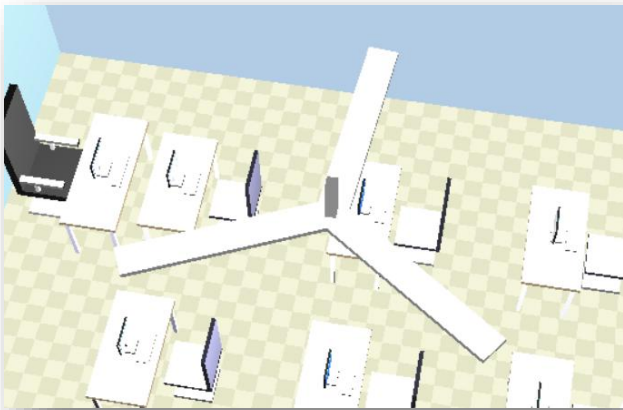


3.6 Fan Animation

Static scenes can be boring, so I added movement. The ceiling fan rotates continuously. I used a timer function that runs every 16 milliseconds (approx. 60 frames per second). This function increases the fanAngle variable, which rotates the fan blades around the Y-axis.

The Code:

```
209 void drawFan() {
210     glPushMatrix();
211     glTranslatef(0.0f, 9.0f, 0.0f);
212     glColor3f(0.9f, 0.9f, 0.9f); glPushMatrix(); glScalef(0.2f, 1.0f, 0.2f); glutSolidCube(1.0f); glPopMatrix();
213     glPushMatrix();
214     glTranslatef(0.0f, -0.5f, 0.0f); glRotatef(fanAngle, 0.0f, 1.0f, 0.0f); glColor3f(0.8f, 0.8f, 0.8f);
215     for(int i=0; i<3; i++) {
216         glPushMatrix(); glRotatef(i*120, 0, 1, 0); glTranslatef(2.0f, 0.0f, 0.0f); glScalef(4.0f, 0.1f, 0.6f); glutSolidCube(1.0f); glPopMatrix();
217     }
218     glPopMatrix();
219     glPopMatrix();
220 }
```



4. Observations and Analysis

During development, I encountered several issues that required problem-solving:

1. **The Door Problem:** Initially, opening the door revealed a solid wall. I fixed this by splitting the wall into three separate pieces, physically leaving a hole for the door frame.
2. **Teacher's Chair Clipping:** The teacher's chair was originally positioned at $Z=-8.0$. When rotated, the backrest went through the wall. I analyzed the coordinates and moved the entire desk forward to $Z=-7.0$ to prevent this collision.
3. **Lighting Brightness:** At first, the room looked washed out and too white. I reduced the Ambient Light settings from 1.0 to 0.4. This introduced shadows in the corners, making the 3D geometry much clearer.

5. Conclusion

This project was a valuable learning experience. I successfully transitioned from simple 2D shapes to building a fully furnished 3D environment. I learned that complex objects are simply collections of basic cubes arranged carefully. I also discovered that good lighting is essential for making a scene look 3D. The final result is a realistic, interactive classroom that fulfills all assignment requirements.